

DIPOLE TYPE BY LENGTH

STEP 0 — always start here $\lambda = \text{wavelength (m)}, l = \text{antenna rod length (m)}, c=3 \times 10^8 \text{ m/s}$

$\lambda = \frac{c}{f} \text{ (m)}$

ratio $\frac{l}{\lambda} \rightarrow$ pick row below

l/λ	Type	Use formula
< 1/50	Hertzian (short)	HERTZIAN $S(R, \theta), D=1.5$
= 1/4	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
= 1/2	Half-wave	half-wave shortcut, $D=1.64, R_{\text{rad}}=73 \Omega$
= 1	Full-wave	ARB, $D=2.41, R_{\text{rad}}=199 \Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

HERTZIAN DIPOLE — $S(R, \theta)$

Far-field E,H phasors small $l \ll \lambda$
 $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta, \quad \vec{H}_\phi = \vec{E}_\theta / \eta_0$
Power density $l/\lambda < 1/50$; far field
 $S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$
 l rod length (m); I_0 peak current (A); R obs. dist. (m); θ from dipole axis; $k = 2\pi/\lambda$ (rad/m); $\eta_0 = 120\pi \approx 377 \Omega$; $S_{\text{max}} = \text{max pwr density (the prefactor)}$.
Max @ broadside $\perp$ to dipole axis = equator plane $\theta_{\text{max}} = \pi/2$; D exact $D=1.5$ (1.76 dB)
Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2 \text{ (W)}$

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R any l
 $S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi l}{\lambda} \cos \theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin \theta} \right]^2$
At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow S = 0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001$ rad $\rightarrow$ tiny $\rightarrow 0$.
Normalized pattern dimensionless $F(\theta) = S(\theta)/S_{\text{max}}$

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion multiply by $\pi/180$ or $180/\pi$
 $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180} \quad \theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$

°	rad	sin θ	cos θ	sin ² θ	cos ² θ
0	0	0	1	0	1
30	$\pi/6$	1/2	$\sqrt{3}/2$	1/4	3/4
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1/2	1/2
60	$\pi/3$	$\sqrt{3}/2$	1/2	3/4	1/4
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY

N identical elements along axis, spacing d , equal phase (broadside, $\theta_{\text{max}} = \pi/2$). Total pattern $S = S_e \cdot F_a$ (S_e = single elem).

Array factor (uniform, broadside) $k = 2\pi/\lambda, \gamma = kd \cos \theta$
 $F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}; \quad F_a^{\text{max}} = N^2$ at $\theta = \pi/2$

Normalize $F_a^{\text{norm}} = F_a/N^2$
HPBW (broadside, uniform) use Nd , not $(N-1)d$
 $\beta \approx 0.88 \frac{\lambda}{Nd} \text{ (rad)} = 50.8^\circ \cdot \frac{\lambda}{Nd}$
Grating lobes appear if $d > \lambda$. Avoid.

BEAM PARAMETERS — β, Ω_p, D

Normalize always start here
 $F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)
Beamwidth β at threshold X
Set $F(\theta_X) = X$, solve for θ_X ; $\beta = 2\theta_X$ (symmetric)

Threshold	$X = 10^{\text{dB}/10}$	θ_X (Gauss. short-cut)
HPBW (−3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
−6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
−10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any X	$F(\theta_X) = X$

Pattern solid angle Ω_p recognize pattern, look up
 $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi \text{ (sr)}$
 $No \phi \text{ dep.}: \Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta.$
 θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0, \pi]$.

Pattern $F(\theta)$	Ω_p (sr)
Isotropic ($F=1$)	$4\pi \approx 12.57$
Hertzian $\sin^2 \theta$	$8\pi/3 \approx 8.38$
Half-wave dipole	≈ 7.66
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta_x^2/2 / \ln 2$
2-axis HPBW (aperture)	$\beta_{xz} \cdot \beta_{yz}$
Other (use integral)	numerical

TI-36X Pro: $[2nd][d/dx]$ for $\int_{\square}^{\square}, \text{RAD mode, multiply by } 2\pi.$

Directivity D always $D = 4\pi/\Omega_p$; common shortcuts below
 $D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$

WORD TRAP: “direction” = θ_{max} (rad); “directivity” = D (unitless).
Dipoles $\rightarrow$ COMMON DIPOLE PARAMS table. Aperture: $D \approx 4\pi A_p/\lambda^2$. 2-axis HPBW: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$. Circular: $D \approx 4\pi/\beta^2$.
Gain (if asked) $\xi = \text{rad. eff.}, 0 < \xi \leq 1$
 $G = \xi D \quad G_{\text{dB}} = 10 \log_{10} G$
Lossless / ideal antenna $\Rightarrow G = D$.

COMMON DIPOLE PARAMETERS

For Friis: $G_t = G_r = D$ below (default lossless $\xi = 1$).

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{\text{rad}} = 36.6 I_0^2 \text{ W}$. Monopole ($\lambda/4$ above ground): same patt. as half-wave but $P_{\text{rad}}, R_{\text{rad}}$ halved.

ANTENNA AREAS (A_e, A_p)

Effective area antenna's RX cross section; D from BEAM PARAMS or table above
 $A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$
Physical cross section wire dipole, diameter d ; l see table above
 $A_p = l d \text{ (m}^2\text{, any dipole)}$
Inverse: D from A_e
 $D = \frac{4\pi A_e}{\lambda^2}$ (unitless)

Thin wire: $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).

FRIIS TRANSMISSION FORMULA

Pwr density at RX, then RX pwr G_t, G_r linear gains; $\lambda = c/f$
 $S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}; \quad P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$
Equivalent: $P_r = S_r A_{e,r}$.
Solve for R max range
 $R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$
Equivalent form using A_e recv. area form
 $P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$
Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.
Off-axis (patterns not aligned): multiply by $F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$, both = 1 when aligned.
Recipe 1. $\lambda = c/f$. 2. Convert any dB gain $\rightarrow$ linear: $G = 10^{G_{\text{dB}}/10}$.
3. Get G_t, G_r by antenna type: dipole $\rightarrow$ COMMON DIPOLE PARAMS; aperture $\rightarrow$ APERTURE box; arbitrary $F \rightarrow$ BEAM PARAMS. 4. Plug into Friis.
Use linear G , not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity ($G, D, P_r/P_t, \text{SNR}, \dots$):

	$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}}$	$X_{\text{lin}} = 10^{X_{\text{dB}}/10}$	
dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: $3 \text{ dB} \Leftrightarrow \times 2, 10 \text{ dB} \Leftrightarrow \times 10$. Add dB = multiply linear.
For E-field / amplitude use $20 \log$ not $10 \log$ (power $\propto E^2$).

NOISE & SNR

Noise power thermal noise into matched receiver
 $P_N = k_B T_{\text{sys}} B \text{ (W)}$
 $k_B = 1.38 \times 10^{-23} \text{ J/K}; T_{\text{sys}}$ system noise temp (K); B receiver bandwidth (Hz).
Signal-to-noise ratio
 $\text{SNR} = \frac{P_{\text{rec}}}{P_N}$ (unitless) $\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$
Typical comm. link needs $\text{SNR} > 10$ log useful signal.

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m^2); A_e effective area (m^2 , $= \lambda^2 D/4\pi$); β_{xz}, β_{yz} HPBWs in two planes (rad); ℓ, d aperture side / diameter (m).

Directivity (any shape, uniform illum.)
 $D \approx \frac{4\pi A_p}{\lambda^2}$ (unitless) $\Rightarrow A_e \approx A_p$

Directivity from beamwidths any aperture
 $D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}}$ (use rad); circular: $D \approx 4\pi/\beta^2$

HPBW and D by aperture shape uniform illum.; β in rad

Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$		$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x \beta_y)$ $= 4\pi \ell_x \ell_y / \lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx 4\pi/\beta^2 = 4\pi \ell^2 / \lambda^2$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx 4\pi/\beta^2 = \pi^2 d^2 / \lambda^2$

Scaling rules $D \propto A_p/\lambda^2; \beta \propto \lambda/\sqrt{A_p}$

Change	New D	New β
Double $(A_p \rightarrow 2A_p)$	area $D_{\text{new}} = 2D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/\sqrt{2}$
Double freq ($\lambda \rightarrow$	$D_{\text{new}} = 4D_{\text{old}}$	$\beta_{\text{new}} = \beta_{\text{old}}/2$

DIPOLE

l — antenna length (m)
 $\lambda = c/f$ (m); $c = 3 \times 10^8 \text{ m/s}$
 I_0 — peak current (A)
 $k = 2\pi/\lambda$ (rad/m)
 R — obs. dist. (m); θ from axis
 $\eta_0 = 120\pi \approx 377 \Omega$
 $S(R, \theta)$ — pwr density (W/m²)

BEAM

$F(\theta) = S/S_{\text{max}}$ (unitless)
 a — coef. on θ^2 in Gaussian
 β — HPBW (rad or °)
 Ω_p — pattern solid angle (sr)
 $R = 4\pi/\Omega_p$ — directivity
 ξ rad. eff.; $G = \xi D$
 $D_{\text{dB}} = 10 \log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIIS / dB

P_t — power transmitted (W)
 P_r — power received (W)
 G_t — transmit gain (linear)
 G_r — receive gain (linear)
 $S_r = G_t P_t / (4\pi R^2)$ (W/m²)
 $P_r = G_t G_r (\lambda/4\pi R)^2 P_t$
 $G = 10^{G_{\text{dB}}/10}$
 $3 \text{ dB} = \times 2, 10 \text{ dB} = \times 10$
 $P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m²)
 $A_e = \lambda^2 D / (4\pi)$ (m²)
 $A_e \approx A_p$ for aperture
 $D \approx 4\pi A_p / \lambda^2$ (unitless)
 $\beta \approx 0.88\lambda/\ell$ (rect/sq.)
 $\beta \approx 1.02\lambda/d$ (circular)
 $2A_p \Rightarrow 2D, \beta/\sqrt{2}$

ARRAY

N — number of elements
 d — spacing (m)
 $\gamma = kd \cos \theta$
 $F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$
 $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$
 $F_a^{\text{norm}} = F_a/N^2$
 $\beta \approx 0.88\lambda/(Nd)$ (broadside)